Set 17: Employee Performance Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse employee performance using a line chart, a department-wise bar chart and a labelled table, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
emp <- data.frame(
  Employee_ID       = c(1, 2, 3),
  Department        = c("Sales", "HR", "Marketing"),
  Years_of_Service  = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
write.csv(emp, "Employee_Performance.csv", row.names = FALSE)  # for Tableau
```

Q1. Line Chart – Performance Trend over Time

> *Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.*

R Program:

```
emp_sorted <- emp[order(emp$Years_of_Service), ]
print(emp_sorted, row.names = FALSE)
cat("Correlation (Years of Service vs Score):", round(cor(emp$Years_of_Service,
emp$Performance_Score), 3), "\n")
p_line <- ggplot(emp_sorted, aes(x = Years_of_Service, y = Performance_Score)) +
  geom_line(colour = "#0072B2", linewidth = 1.1) + geom_point(size = 4, colour = "#0072B2") +
```

```
  geom_text(aes(label = paste0("Emp ", Employee_ID, " (", Department, "): ",
Performance_Score)), vjust = -1.2, size = 3.6) +
  scale_x_continuous(breaks = 2:8, limits = c(2, 8)) + scale_y_continuous(limits = c(70, 100))
+
  labs(title = "Employee Performance Trend over Years of Service", x = "Years of Service", y =
"Performance Score") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_line)
ggsave("s17_q1.png", p_line, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Employee_ID Department Years_of_Service Performance_Score
           2         HR                3                92
           1      Sales                5                85
           3  Marketing                7                78
Correlation (Years of Service vs Score): -1
```

Output:

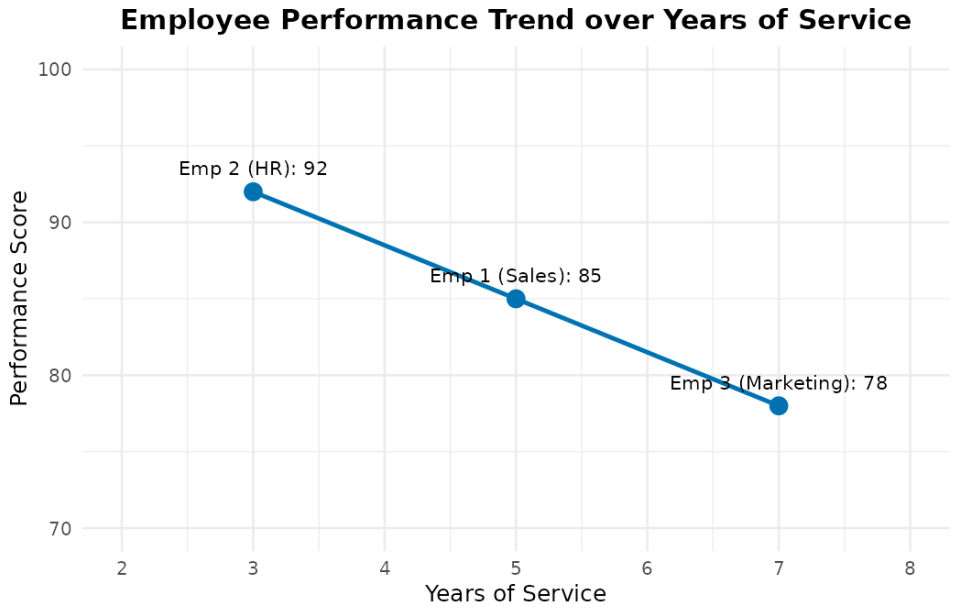


Figure 1: Performance score across years of service

Inference:

The dataset has no dates, so **Years of Service** is used as the time axis. Performance falls steadily as service increases: 92 (3 years, HR), 85 (5 years, Sales), 78 (7 years, Marketing). This is a perfect negative correlation (r = −1).

With only three employees, each from a different department, this compares employees rather than tracking one person over time. More records would be needed to confirm a real trend.

Q2. Bar Chart – Employees across Departments

> *Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.*

R Program:

```
dept <- as.data.frame(table(emp$Department)); names(dept) <- c("Department", "Employees")
print(dept, row.names = FALSE)
p_bar <- ggplot(dept, aes(x = Department, y = Employees, fill = Department)) +
  geom_col(width = 0.55) + geom_text(aes(label = Employees), vjust = -0.5, size = 4.5) +
```

```
  scale_fill_manual(values = c(HR = "#009E73", Marketing = "#E69F00", Sales = "#0072B2")) +
  scale_y_continuous(limits = c(0, 1.5), breaks = c(0, 1)) +
  labs(title = "Distribution of Employees across Departments", x = "Department", y = "Number
of Employees") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_bar)
ggsave("s17_q2.png", p_bar, width = 7, height = 4.5, dpi = 150, bg = "white")
```

Console Output:

```
 Department Employees
         HR         1
  Marketing         1
      Sales         1
```

Output:

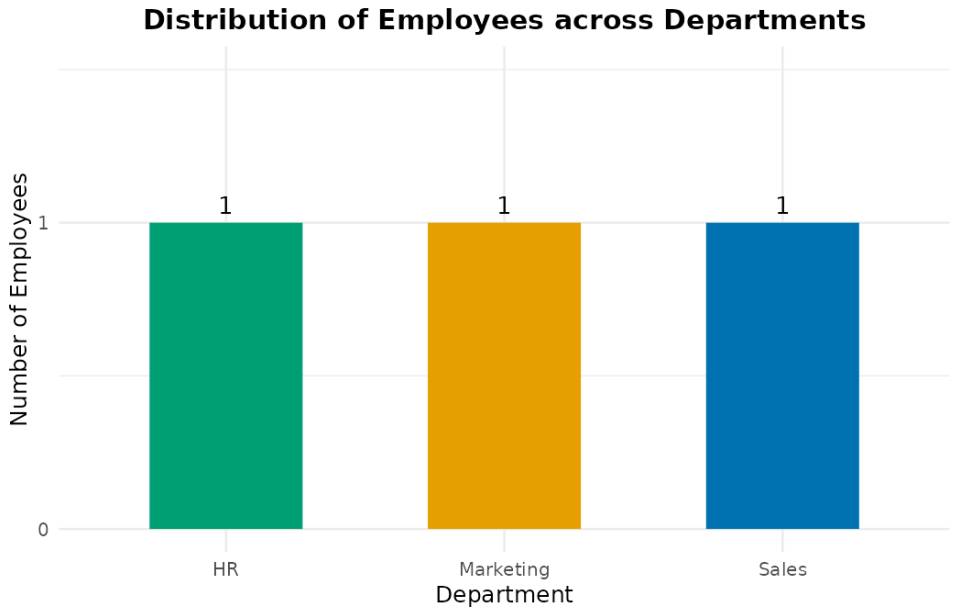


Figure 2: Number of employees per department

Inference:

Each department (HR, Marketing, Sales) has exactly one employee, so the workforce is evenly distributed.

Q3. Table – Performance Data for Each Employee

> *Build a table to display the performance data for each employee. Label the table elements.*

R Program:

```
tbl <- emp
names(tbl) <- c("Employee ID", "Department", "Years of Service", "Performance Score")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = tbl_theme)
png("s17_q3.png", width = 1000, height = 260, res = 150)
grid.arrange(grid::textGrob("Employee Performance Data", gp = grid::gpar(fontsize = 14,
fontface = "bold")),
             tg, ncol = 1, heights = c(0.3, 1))
invisible(dev.off())
```

Console Output:

```
 Employee ID Department Years of Service Performance Score
           1      Sales                5                85
           2         HR                3                92
           3  Marketing                7                78
```

Table Output:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Inference:

Employee 2 (HR) has the highest score (92) with the fewest years of service (3), and Employee 3 (Marketing) has the lowest score (78) with the most service (7).

Q4. Tableau Dashboard – Interactive Exploration

> *Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.*

R Program:

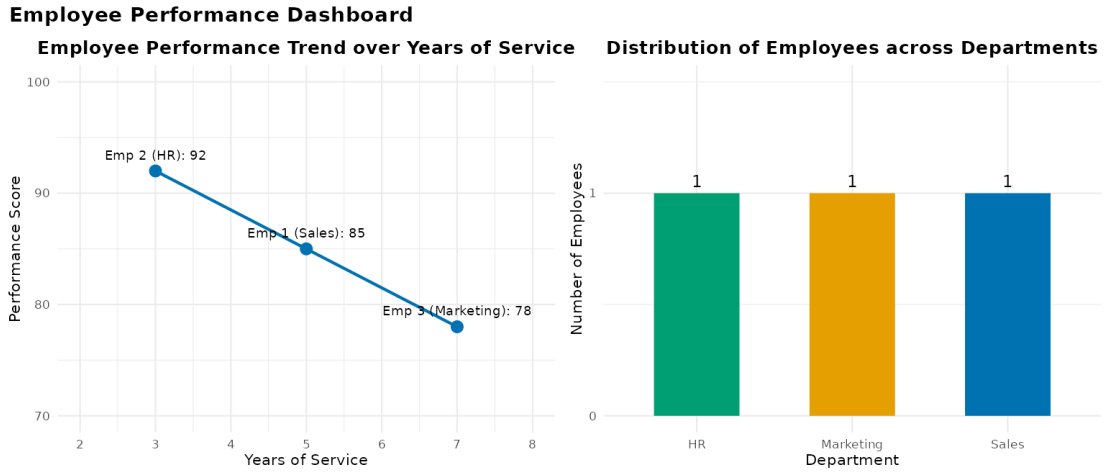
```
dash <- (p_line | p_bar) / wrap_elements(full = tg) + plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Employee Performance Dashboard",
                  theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s17_dash.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Employee_Performance.csv**. Convert **Employee ID** to a Dimension.
2. **Sheet 1 – "Performance Trend":** drag Years of Service (as **Dimension, Continuous**) to Columns and AVG(Performance Score) to Rows. Set the Marks type to **Line** with markers. Drag Department and Employee ID to Label. Title the sheet and axes.
3. **Sheet 2 – "Department Distribution":** drag Department to Columns and **CNT(Employee ID)** to Rows. Drag Department to Color and show mark labels.
4. **Sheet 3 – "Performance Table":** drag Employee ID, Department, Years of Service and Performance Score to Rows (discrete), or use Measure Names / Measure Values. Rename the headers.
5. From the menu choose **Dashboard → New Dashboard** and set **Size → Automatic** (or Fixed 1200 × 800).
6. Drag the sheets onto the canvas: Performance Trend (top-left), Department Distribution (top-right), Performance Table (bottom). Add a **Text** object at the top as the dashboard title.
7. On each chart, open the sheet's drop-down and select **Use as Filter**, so clicking a bar, point or slice filters the other views.
8. Go to **Dashboard → Actions → Add Action → Highlight** (Run on: Hover) so related marks highlight across all views.
9. Right-click each quick filter → **Apply to Worksheets → All Using This Data Source** (Employee_Performance) so one filter controls every view.
10. Add a **Department** quick filter (multiple values list) to the dashboard.

11. Save as a packaged workbook (**File → Save As → .twbx**) or publish to Tableau Public.

Dashboard Layout:



Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a department bar highlights that employee on the trend line and filters the table, linking each employee's department, tenure and score.

Result

The employee performance data was visualised with a line chart, bar chart and labelled table in R, and combined into an interactive Tableau dashboard. In this sample, performance decreases as years of service increase (92 → 78), and employees are evenly spread across departments.